

Research Paper

Design and Development of Real-time Code Editor for Collaborative Programming

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Abstract— The Internet is expanding quickly, and many desktop apps are already being developed for the online. Using the Internet, many apps may be easily accessible at any time and from any location. Code editors are one of the tools that developers require to design their applications. The goal of this research is to employ web socket technology to build and develop a real-time code editor application that facilitates user collaboration while working on the project. Users of this program can work together in real time on a project thanks to its capability. The authors employed an analysis process that includes studying the literature, and studying the code editor software that are currently in use. CrossCode is a web application that offers a workspace for creating, performing, and collaborating with other users in real-time over the terminal. The primary functionalities of the application include a workspace for creating, running, and building source code as well as real-time chat and terminal building. The programming languages C, C++, Python, and Java are supported by this application. The current research contributes to the advancement of online code collaboration tools, providing developers with an efficient and accessible platform for remote teamwork and code sharing.

Keywords—Real-time Code Editor, Web Socket Technology, Collaborative Coding, Multi-language Support, Cloud-based Code Development

1. Introduction

Software engineering has seen a trend of technical advancement as software design started to shift from desktop to web platforms. Even though there are a lot of IDE (Integrated Development Environment) programs available these days, including Eclipse and Visual Studio, desktop IDEs still have a lot of drawbacks, like the requirement to install plug-ins and take a long time to configure before using the program can run [1]. In cases when numerous devices require configuration, this issue may prove to be a significant time waster [2].

Numerous software programs are operated on the cloud and employ a web browser as their user interface, which enables instantaneous collaboration, universal access, and eliminates the need for desktop computer installation and configuration [2].

Single IDE (like pair programming) is one of the technologies used for instantaneous collaboration. The technique of letting two programmers' access and work on the same code in a

single development environment is known as pair programming [3]. Programmers can develop, edit, and remove source code in real time when working in pairs. Program code synchronization issues could be resolved with pair programming, which would allow any programmer working on the same project to see who made changes to the code and observe when they happen.

Programmers working in different places but in the same environment to discuss or fix flaws in the program could benefit from collaborative technologies. Thus, it must develop an application that enhances program creation performance, such as real-time collaboration, and use a terminal to build, run, and display the program's output.

1.1 Real-time Communication

Instant messaging and voice-over-IP (VoIP) telephony are two examples of communication technologies that are paired with a variety of collaborative applications to provide real-time communication (RTC) [4]. The four interconnected building components that make up RTC technologies are contextualization, presence awareness, unified communications, and the E-Collaboration portfolio [5].

RTC systems often facilitate multi-person conferences and two-person interactions. RTC systems enable real-time collaboration between users by integrating several synchronous communication mediums into an integrated environment [6]. This includes editing documents, making voice calls, and hosting multi-person video conferences. RTC technology may be able to assist in resolving the synchronization issue, particularly in collaborative settings. The RTC systems' building blocks are presented in Table 1. RTC systems provide users with freedom and the ability to engage with others remotely. They have also improved and facilitated interaction and communication, which could help in the development of the project more successfully.

Table 1. Building Blocks of Real-Time Collaboration Systems

Unified Communication (UC)	Presence Information Portfolio	Contextualization	E-Collaboration
• Various media and communication channels • Media and device integration • Rule-based configuration of message routing and call diversion • Definition of preferred media • Unified messaging portal	• Signaling of people and communication channel • Aggregation of presence information on group, role, and objective level • Active management of buddy list. • Individualized and automatic signaling	• Integration with applications • Context specific buddy lists • Mobile RTC with location-based services	• Audio and video and conferences seminars • Adhoc application sharing • Team calendars and contact management • Joint whiteboards and discussion forums • Document folders

1.2 State of the Art

The goal of this research is to develop a code editor application that can be used for collaborative work on a particular file, syntax checking, source code building and running through a terminal, and chat as a medium for communication between users. This application offers a workspace with handy features including auto complete in C, C++, Python, and Java, as well as the ability to generate, edit, run, and build source code that has already been written. The upload and download features can be used by users who wish to import project files from the desktop or export source code files to their desktop. This application may be able to improve project development performance and assist programmers in working together on projects in real-time.

The rest of the paper is organized as follows: Section 2 provides a comprehensive literature review, covering the historical evolution of collaborative tools and examining key features, communication protocols, and challenges in real-time code editors. Section 3 details the research methodology and design principles employed in this study. Following this, Section 4 delves into the implementation phase, providing insights into the technical aspects and challenges faced. Section 5 then presents the results, opening the door to a robust discussion in Section 5 that connects our findings with existing literature and explores the implications. The paper

concludes in Section 6, offering a concise summary of key contributions and suggesting avenues for future research in collaborative programming tools.

2. Related Work

The landscape of collaborative programming has evolved rapidly in response to the increasing demand for efficient teamwork in software development. This literature review explores existing research and developments in the design and development of real-time code editors, focusing on their role in facilitating collaborative programming endeavors. A Problem-based Collaborative Learning Groupware for improving computer programming skills is a software platform or tool designed to facilitate collaborative learning experiences centered around solving programming problem [7]. Author [8] presented a design of a modern online interview platform for a recruitment system involves integrating various features to streamline the hiring process, improve candidate experience, and provide valuable insights to recruiters. Collaboration tools are essential for developers working on projects, whether they are part of a small team or a large, distributed development environment [9]. Collaborative programming can positively impact students' achievement in learning Object-Oriented Programming by fostering a supportive learning environment, enhancing critical skills, and providing a more holistic understanding of OOP concepts [10]. Improv [11] is an IDE extension for preparing and delivering code-based presentations. Authors [12] presented the challenges faced by teachers during online classes and to assess their readiness for blended learning. Authors [13] presents results of study conducted to explore and uncover the proficiency level of Information and Communication Technology (ICT) among undergraduates in public universities in Nigeria.

2.1 Real-time Collaborative Programming

When working on development projects, any programmers are working in teams. Within the same project file, any programmer with access to the code can add, edit, and create new code. Thus, synchronization amongst programmers is necessary to prevent code duplication, and integrated real-time collaboration in a single environment is essential to tackle this synchronization problem.

The purpose of the Integrated Development Environment (IDE) is to provide programming teams a collaborative environment where they can run and build code, edit text in real time, communicate, and do a lot of other things. Multiple users can collaborate while editing a document and instantly show the changes to other users who have access to the same document because of the ability to edit text in real-time. Numerous free programs, including Google Docs, offer real-time text editing capabilities. This functionality can be quite useful in programming in addition to providing great collaboration for everyday users.

Additionally, there is a vast array of web-based technologies available for collaboration. Like EtherPad, which enables in-the-moment text editing. Web-based text editing tools like

Ace and CodeMirror are intended to be integrated into IDEs and applications.

Collaboration tools are highly helpful in increasing the efficiency of project creation, as software development projects necessitate coordination and cooperation amongst programmers [14]. The efficiency of teamwork in programming can raise the standard and productivity of a project or segment of software.

Real-time collaborative programming enables programmers to work together on a single programming file. When a programmer types code, the real-time system [15] will automatically integrate it without the programmer's manual intervention. The same source code directory can be accessed and edited by many programmers, even simultaneously.

In real-time sessions, programmers can work together by entering and exiting real-time sessions during collaborative programming. When a new client wants to join, there are procedures to follow in order to use the connection protocol with two-way client to enter and exit the session.

A distributed join protocol was designed with the following message to support the new client request:

- JOIN: A new client can request to join an ongoing real-time session by sending a request to the session manager.
- START: The session manager notifies all current clients that a new client request has been received, initiating the join-protocol procedure.
- READY: The session manager receives information from all current clients regarding their readiness to enter the state.
- FINISH: The session manager notifies all of the current clients that all procedures have been completed.



Figure 1. Join Protocol in Real-Time [15]

2.2 Integrating Collaborative Program Development and Debugging in a Virtual Environment

Programmers use IDEs (Integrated Development Environments) as frequently as any other tool. The IDE was developed to facilitate real-time collaboration in software development and engineering, allowing programmers to collaborate on the same project in real-time while designing, discussing, and sharing it.

Tasks related to programming can be shared by programmers in an integrated collaborative environment. ICI (Idaho Collaborative IDE) is one of the tools used to implement collaborative IDE [16]. Programmers can work collectively in real-time on a variety of software development tasks even if they are located in distinct locations because of ICI. ICI

integrates a real-time debugger with synchronous collaboration technology.

Collaborative technologies are being implemented in the field of education and programming. In order to facilitate real-time cooperation throughout the compilation, linking, running, and debugging of sessions, remote computer science education requires a more collaborative IDE integration, which can develop a platform that makes textual or voice communication amongst developers easier.

Programmers can interact with one another and monitor what other programmers are doing in the same space when they work in a collaborative setting. Programmers can communicate in real time with other programmers or project teams using text or voice chat. Programmers can ask one another to participate in a collaborative IDE session, where they can work collaboratively on program design, coding, or debugging within the same development environment [16].

2.3 Real-Time Collaborative Coding in a Web IDE

IDEs have undergone a significant transformation, evolving from solitary coding tools to collaborative platforms. Their role has expanded to not only aid individual project development but also facilitate seamless communication and collaboration among programmers. A notable example is Collabode, an IDE specifically designed for real-time collaboration. This shift towards collaborative IDEs reflects a recognition of the growing significance of teamwork in optimizing the efficiency of the software development process.

Collabode is an integrated development environment (IDE) for Java that facilitates teamwork online. When several users make modifications with Collabode, the results are instantly shared with other users without the need for manual programming [14]. Additionally, multiple users can access this application at once within the same project module.

Collaboration is conducted with the aim of enhancing the quality of the project output and collaboration. Users simply need to access the same URL as other users in order to collaborate on projects. Collabode facilitates editor collaboration by using EtherPad. Collabode uses Eclipse to manage the projects, offering features like syntax highlighting, continuous compilation, compile errors and warnings, code formatting and refactoring, and program code execution.

For restricted synchronous collaboration, three techniques are utilized: Mobile Instructor, Micro Outsourcing, and Test Driven Programming [17]. The three models listed above show how differently programmers can collaborate with one another. They all use different approaches, like side-by-side programming, pair programming, and other collaborative methods.

3. Methodology and Design

In this section, the methodology and design aspects of the study is presented. Section 3.1 explores the CrossCode

Features Algorithm, while Section 3.2 delves into the Operational Transform Algorithm. These algorithmic frameworks serve as pivotal components in the overall architecture and functionality of the study, shaping the foundation for further exploration and analysis.

3.1 CrossCode Features Algorithm

CrossCode integrates with social networks to facilitate collaboration and uses the ReactJS plugin for login to make using the application easier. Users can select Java, C, C++, Python, and C as their primary programming languages in CrossCode. To view the code program's results, users can run it and develop the program. The generated parent folders and the source code program are available for download by users. Users can work together in real-time to build projects by collaborating. Chat serves as a medium for communication amongst users who are part of the same project. Users can email the website with questions or to report any issues that they find.

The workspace screen appears as soon as a user logs into the program successfully. The algorithm shown in Figure 2 describes how all completed projects, folders, and files—if any—will be displayed after the user logs in.

```

1: //javascript
2:  $userId \leftarrow REQUEST['userId']$ 
3: send  $userId$  with  $ajax$ 
4: //php
5: function getProject( $userId$ ) // function getProject on the service
   with  $userId$  parameter
6:   load_database
7:   get project from database Project based on  $userId$ 
8: end function
9: function getFolder( $userId$ ) // function getFolder on the service
   with  $userId$  parameter
10:  load_database
11:  get folder from database folder based on  $userId$ 
12: end function
13: function getFile( $userId$ ) // function getFile on the service with
    $userId$  parameter
14:  load_database
15:  get file from database file based on  $userId$ 
16: end function
17: //javascript
18: if success then // if request was success
19:   view project, folder, file
20: end if

```

Figure 2. Algorithm to view all project, folder, and files

The project is displayed in the workspace's file directory on the left. The file directory will appear empty if the user is new. Real-time connections are provided by this program, ensuring that all changes are updated simultaneously. Thus, this application has the auto-saving feature depicted in Figure 3, which will store the code automatically if the user doesn't touch the keyboard for a minute.

```

1: //javascript
2: initialisation  $flag \leftarrow false$ 
3: initialisation  $flagevent$ 
4: if  $keyup()$  then // Run when button was not pressed
5:   if  $flagevent \neq undefined$  then
6:     clearTimeout  $flagevent$  // Clear timeout on
        $flagevent$ 
7:   end if
8:    $flagevent \leftarrow function settimeout 60$  // Set the time
       of timeout along 60 seconds
9:   if  $flag = true$  then
10:     $code \leftarrow REQUEST['code']$ 
11:     $fileId \leftarrow REQUEST['fileId']$ 
12:    send  $code$  and  $fileId$  with  $ajax$ 
13:  end if
14:   $flag \leftarrow true$ 
15: end function
16: end if
17: if  $keydown()$  then // Run when button was pressed
18:    $flag \leftarrow false$ 
19: end if
20: //php
21: function updateFile( $fileId$ ,  $code$ ) // function updateFile which
   on the service with  $fileId$ , and  $code$  parameter
22: update  $code$  from file database with  $fileId$  based on
   parameter
23: end function

```

Figure 3. Algorithm to auto saving code

The algorithm to run and build program is depicted in Figure 4. The user will view the outcome after creating the project file and running the software. The findings will be shown on the website as a terminal. The error message will appear on the terminal in the event that an error occurs.

```

1: //javascript
2:  $textarea \leftarrow REQUEST['textarea']$ 
3: file Type  $\leftarrow REQUEST['fileType']$ 
4: if  $f5$  pressed then
5:   send  $textarea$ , and  $fileType$  with  $ajax$  // send to function which on the service
6: end if
7: //php
8: function code ( $textarea$ ,  $fileType$ )
   // function code which on the service with  $textarea$ , and  $fileType$  parameter
9: file write  $\leftarrow text$  area // create file contain text area from parameter
10: file extension  $\leftarrow filetype$  // create file extension based on  $textareaType$  parameter
11: compile file // compile that file with compiler based on their extension
12: view outcome from file compile
13: end function
14: //javascript
15: if success then // if request was successa
16:   open terminal data // open terminal contain data which it has been compiled
17: end if

```

Figure 4 Algorithm to run and build program

When creating a project, collaboration capabilities greatly aid in enhancing communication between the programmer and performance [19]. Our system includes collaboration features, as seen in Figure 5, where friends can be invited once they create a project file. By imposing access limits, the project owner will forward the request to another user who wishes to get an invitation to join. in order for other users to act in accordance with the access given.


```

1: run term.js server in nodeJS
2: fork terminal
3: run compile.sh
4: //javascript
5: build terminal
6: create new pid from term.js
7: function viewTerminal(pid)
8:     kirim pid //terminal was viewed
9: end function

```

Figure 5. Algorithm to collaborate among different users

The algorithm used to create and display terminals are shown in Figure 6. User can see terminals by hitting the run button or the F5 key on the keyboard. The NodeJS server will launch when the button is hit, causing the terminal to open in a new tab. The error that is present in the code can also be seen at the terminal.

```

1: //javascript
2: friendId ← REQUEST['friendId']
3: userId ← REQUEST['userId']
4: accessRestriction ← REQUEST['accessRestriction']
5: send accessRestriction, friendId, and userId with ajax
6: //php
7: function insertCollaborate(friendId, userId, accessRestriction)
    // function insertCollaborate on the service with friendId,
    // userId, accessRestriction parameter
8: get in all parameters to file database
9: end function

```

Figure 6. Algorithm to build and display terminal

3.2 Operational Transform Algorithm

A group of users can access and modify the same document simultaneously over a network connection via real-time collaboration. Because every user action will be carried out immediately and without delay, consistency of data must be retrieved simultaneously in order to connect in real-time. Operational transformation algorithms are employed in collaborative real-time groupware to guarantee consistency of copies of shared data.

One technique for ensuring consistency among copies of shared data on group editors is operational transformation (OT) [20]. The integration algorithm and transformation functions are the two primary components, which are clarified by the operational transformation method [21]. The process is carried out, transmitted, and received by an integration algorithm. The transformation function, on the other hand, is defined on the same state and is used to combine two concurrent processes.

Operational transformation would subsequently be carried out by receiving operations in accordance with local concurrent operations [21]. The word processor program is one example of how the operational transformation method is implemented. A word processor application may perform extensive editing functions and has a sophisticated structure

[22]. Word processing software not only allows for the creation and deletion of objects, but it also allows for the modification of an object's attributes. The ability to update collaboratively is crucial for collaborative word processing applications as it expedites the development of documents. Extension operational transformation is utilized to support the update operation's concurrent execution. This approach, for instance, has led to the creation of the CoWord project, which attempts to transform Microsoft Word into a real-time collaborative word processor without modifying the original code.

The two layers that make up the component of operational transformation are the low-level transformation functions and the high-level transformation control algorithms [22]. The approach taken in the OT extension is to incorporate a new update-related transformation function while leaving the high-level control algorithm unaltered. Localizing the extension and reducing complexity are made possible in this approach.

The update operation differs from the insert and delete operations in that the former alters the document's linear address space while the latter does not. Insert remained unchanged when it converted to a concurrent update. The Update's position will, however, be increased by one when it is transformed to Insert if its position is more than or equal to the Insert position.

4. Implementation

This section discusses the implementation of CrossCode. To access the workspace, a user must first log in. The user's data will first be examined to see if it has ever been kept in the database. New log data will be created in the user, file, and log collections if the data already exists, else it will be recorded into the collection logs. In addition, the user will be taken to the workspace page, where project file data will be loaded from the user's file database and collection. After the user is redirected to the workspace, their Facebook friends list will load. The friend's list will be utilized to invite new collaborators, and those who accept invitations will be able to speak with other individuals.

The architecture is shown in Figure 8 when a user initiates a chat with another user. The JSON data is sent when the user types a message and hits the enter key, and the chat data is then saved in the chat collection. Subsequently, the information is transmitted to the socket.io server and subsequently distributed to all users that were previously invited. How to add a collaborator to join a project using the same file is shown in Figure 9. To work together with other users, the user can look for friends' names who have signed up for CrossCode. Data will be entered into file collection and sent to the socket.io server after the user ID has been obtained. Lastly, the collaborator will be added to the project's same file.

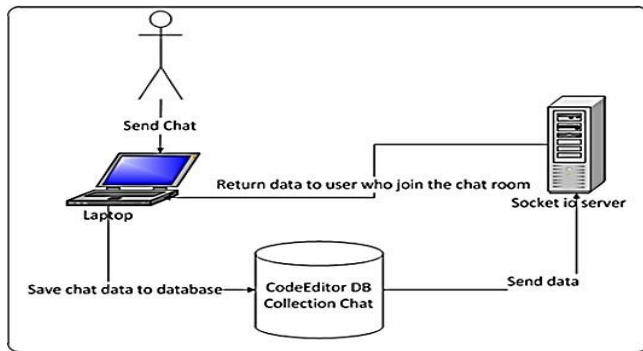


Figure 8. Chat architecture

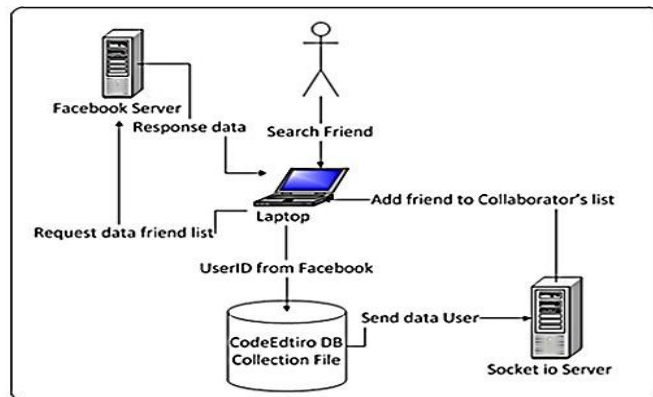


Figure 9. Add collaborator architecture

To access the folders and files in the workspace, users can click the button located in the left corner of the page. A tree-shaped display of the folders and files will pop up.

As shown in Figure 10, this application allows for real-time user participation. This application allows for real-time user participation. Using the friend list located in the box on the right, the user can extend an invitation to other users to work together on the same project. The user can chat with another user after getting an invitation to join them in the same project. A person can invite pals who are on their friend list if they are signed in using their Facebook account. Users will automatically connect with other users who have utilized anonymous accounts when they log in. Figure 10 shows a chat window with the user's name, the date, and a search option to locate friends who would like to be asked to collaborate or communicate via chat.

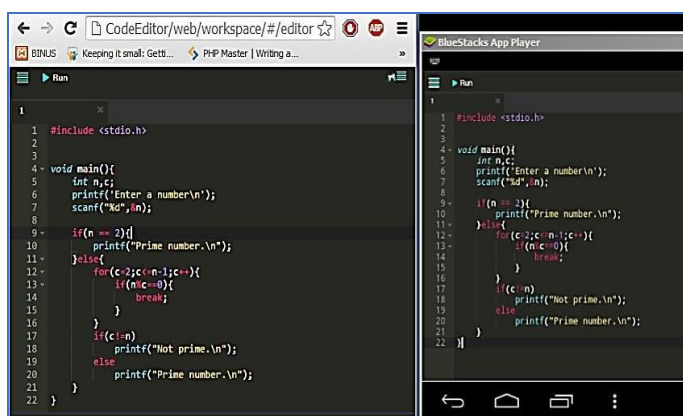


Figure 10. Collaboration between web and mobile user

5. Results and Discussion

This research employs the Apache Benchmark, commonly known as ab (Apache HTTP Server Benchmarking Tool), as a robust tool for assessing the performance of our application. Apache Benchmark is recognized for its proficiency in measuring the performance of the Apache Hypertext Transfer Protocol (HTTP) server. In this study, we leverage ab to conduct a thorough evaluation of our application, scrutinizing its responsiveness, scalability, and overall efficiency. The number of requests per second that Apache can handle is displayed by the Apache Benchmark. RAM is the kind of hardware that will have the most significant effect on the webserver's performance.

Primarily, Apache Benchmark was capable of doing load testing on a webpage in both concurrent and sequential modes. Additionally, it could test the results in terms of time spent on each request, amount of concurrency, transfer rate, unsuccessful requests, total transfers, test duration, finished requests, errors in writing, HTML transferred, and requests per second. Numerous options are available with Apache Benchmark, including -n for number of requests for benchmarking sessions and -c for number of requests to be managed concurrently.

This examination involves sending 1000 requests to the specified URL with a maximum concurrency of 1000 simultaneous requests. The tests are carried out on two distinct computer configurations, each defined by specific specifications. For SPEC 1, the system is equipped with a CPU Pentium Dual Core E2180 and 4 GB of RAM. On the other hand, SPEC 2 features a Core i7 2600 processor and is outfitted with 8 GB of RAM. The experimental results are presented in Table 2.

Table 2. Results of experimentation

	SPEC 1	SPEC 2
Total data transferred for 1000 requests (in Bytes)	3575000	3501000
Test Completion Time (in Seconds)	28.842	0.442
Requests per seconds	34.67	2264.06
Time per request (in ms)	28842	0.442
Transfer rate (in Kbytes/sec)	121.05	7740.70

Apache Bench reveals the average load time (in ms) of the 1000 request. The Apache Bench results for the first test reveal that the average load time for the 1000 requests is 14.6 seconds (14,608 ms), and longest load time as 23.8 seconds (23,388 ms). In the second test, which focuses on connection time the average load time is remarkably shorter at 0.078 seconds (78 ms), whereas longest load time is 0.335 seconds (335 ms). These results provide insights into the application's performance under different testing conditions.

An in-depth analysis of the Apache Benchmark findings provides insightful information about the evaluated application's overall efficiency, scalability, and responsiveness. These findings, along with the apparent hardware-dependent performance differences between SPEC 1 and SPEC 2, offer a more detailed understanding of the factors influencing the performance landscape of the application. These judgments can serve as a basis for further investigation and improvement that could optimize the

program and improve user experience while encouraging more efficient use of resources. This iterative process emphasizes how crucial it is to continuously evaluate and modify the application in order to bring it into compliance with changing user expectations and performance goals.

6. Conclusion and Future Scope

The design and execution of the real-time collaboration application are presented in this paper. CrossCode is an web application that facilitates real-time collaboration between programmers via chat or invites them to join a project, as well as project management functions like import, export, and shared projects. It also allows programmers write code and view the results of that code when it is performed. The programming languages C, C++, Python, and Java are supported by CrossCode. The primary functionalities of CrossCode include a workspace for creating, running, and building source code as well as real-time collaboration, chat, and terminal building. The authors will recommend a few features for more study and development, such as the ability to automatically generate UML designs from written code. At current state, CrossCode can only edit a single file. It would be helpful if the program code could be edited in several files in the future version. This way, the user could replace all of the code that is located in the same project in one go if they searched for a specific term. CrossCode might potentially be developed into a repository-like system, similar to GitHub, for organizing both created and ongoing projects.

Conflict of Interest

There is no conflict of interest.

Funding Source

None

Authors' Contributions

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Mohammad Bilal Khan, a dedicated B. Tech. (CSE) student at Prestige Institute of Engineering Management and Research, Indore. He is a skilled programmer and adept at leading a team. He actively engages in research endeavors with a primary focus on Computational Intelligence.



Chandra Shekhar Kushwaha, a dedicated B. Tech. (AI & DS) student at Prestige Institute of Engineering Management and Research, Indore, excels in coding and collaborative teamwork. Actively engaged in research, his focus lies on Artificial Intelligence. Chandra Shekhar is a committed student who consistently contributes as a valuable team member.



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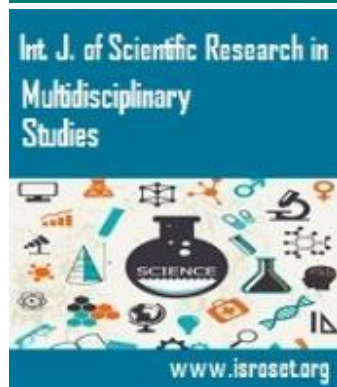
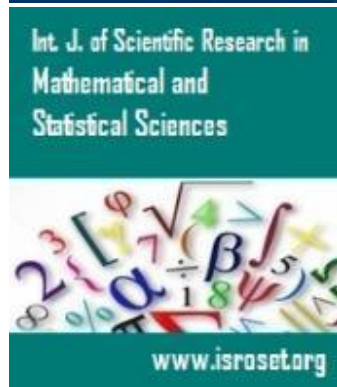
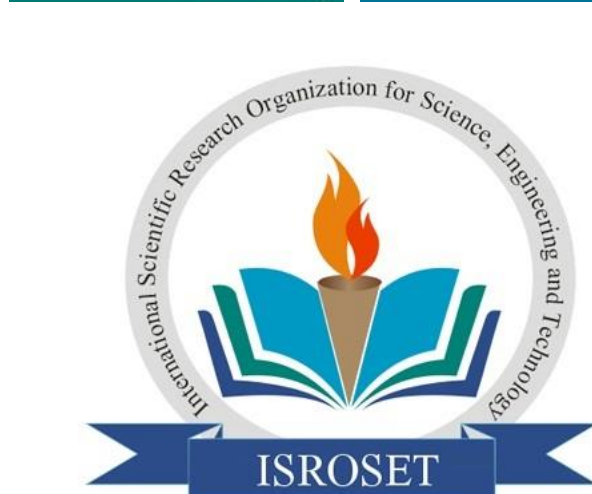


Avni Verma, a dedicated B. Tech. (AI & DS) student specializing in Artificial Intelligence and Data Science at Prestige Institute of Engineering Management and Research in Indore. Avni showcases versatile skills as a proficient content writer and possesses expertise in web development.



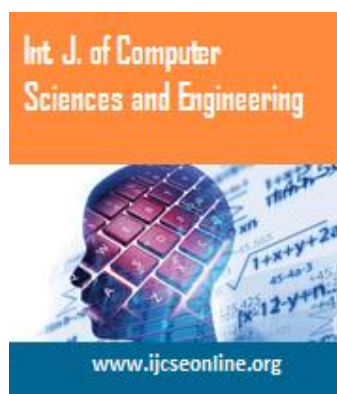
Dr. Pritika Bahad is an experienced academician and researcher with a Ph.D. (Computer Science) and M.Tech. (Mobile Computing Technology) from Devi Ahilya University, Indore. She is currently working as Assistant Professor in Department of Artificial Intelligence & Data Science at Prestige Institute of Engineering Management and Research, Indore; she brings 15 years of rich teaching experience and research expertise.





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